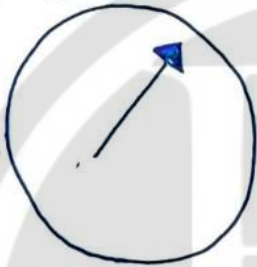


OSCILLATION

⇒ Periodic motion = when a body repeats its motion in a definite time interval on a definite path then its motion is called periodic motion)

Example = Earth, completes one round, around the Sun in one year

→ Second hand of watch doing circular motion



⇒ Non-periodic motion = when a body does not repeat its motion in given time interval then its motion is called non periodic motion.

- walk of human being
- motion of vehicle on straight road.
- Study of backloggers student

⇒ Oscillatory = The to and fro motion of an object from its mean position is called oscillatory motion.

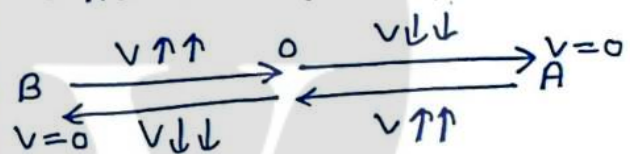
→ Example =

[motion of pendulum of wall clock, motion of a mass hanging on a spring. etc.]



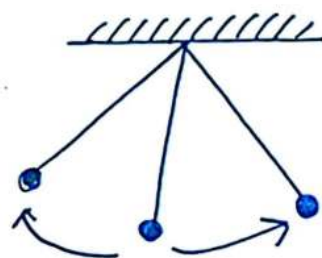
→ At Extreme position $V=0$

→ At mean position $= V_{max}$



⇒ Difference between oscillatory motion and vibratory motion =

- oscillatory motion at low frequency
- vibratory motion at high frequency



Oscillatory motion

⇒ Simple Harmonic motion = (S.H.M.) "The oscillatory motion in which the displacement of particle is straight line".

→ It is a simple form of oscillatory motion.

→ Oscillatory motion with small amplitude is called S.H.M.

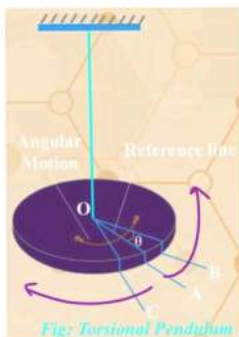
→ To have motion in straight line

Restoring force = The force which brings the object towards mean position

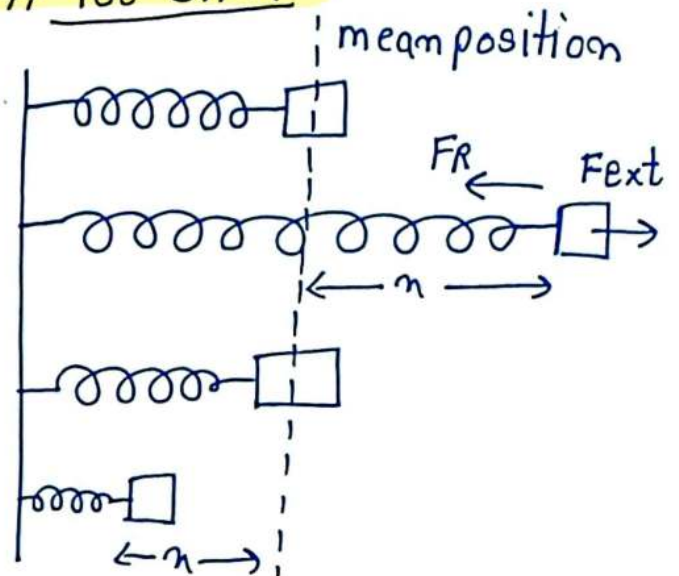
(1) Linear Simple harmonic motion = path of motion is straight line.



(2) Angular Simple Harmonic = motion path of motion is an arc of circle



for S.H.M. =



→ Let a mass of m is attached with a spring. On applying force the displacement from mean position x

$$F_R \propto x$$

$$F_R = -Kx \dots (i)$$

and $F = ma \dots (ii)$

from Equation (i) and (ii)

$$ma = -Kx$$

$$a = -\frac{K}{m}x$$

where =

$$\omega^2 = \frac{K}{m}$$

$$\Rightarrow a = -\omega^2 x$$

$$\Rightarrow \boxed{a \propto -x}$$

In the above condition there will be S.H.M.

→ $\vec{F} = -K\vec{x}$ [force is always towards mean position]

→ ' x ' displacement from mean position

→ k is Restoring force constant

⇒ Differential Equation of linear S.H.M. = Let a mass

m is attached with a spring and displacement from mean position is x

$$F = ma \dots \dots (1)$$

And $F = -Kx \dots \dots (2)$

from equation (1) and (2)

$$ma = -Kx$$
$$a = \frac{-K}{m} x \quad \left[\omega^2 = \frac{K}{m} \right]$$

$$a = -\omega^2 x$$

$$\frac{d^2 x}{dt^2} = -\omega^2 x$$

$$\boxed{\frac{d^2 x}{dt^2} + \omega^2 x = 0}$$

This is differential equation of S.H.M and General solution of this equation

$$\boxed{x = A \sin[\omega t + \phi]}$$

→ Similarly for angular S.H.M.

$$\tau \propto \theta \quad \left[\begin{array}{l} \tau = \text{Torque} \\ \theta = \text{angular} \end{array} \right]$$

$$\frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0$$

$$\theta = \theta_0 \sin(\omega t + \phi)$$

$$\boxed{\begin{array}{l} x \rightarrow \theta \\ F \rightarrow \tau \end{array}}$$

$$\boxed{x = A \sin[\omega t + \phi]}$$

x = displacement from mean position

A = Amplitude max displacement

ω → angular frequency

ϕ → Epoch or Initial phase angle

⇒ Some important definition related to S.H.M. =

(i) Mean position = The position at which the restoring force on the particle is zero.

(ii) Instantaneous displacement = At any time t the displacement of a particle moving in simple harmonic motion from the mean position is called Instantaneous displacement.

(iii) Amplitude (A) = The maximum displacement from the equilibrium position of a particle moving in simple harmonic motion is called Amplitude

(iv) Time period (T) = The time taken in one oscillation or vibration

(v) Frequency (n) := The number of oscillations or vibrations per second is called frequency $n = \frac{1}{T}$

⇒ Some Important definition related to S.H.M. =

(vi) Angular frequency (ω) = The change in phase angle of a particle per second is called angular frequency.

(vii) phase angle = The quantity by which at any moment the position or state of motion of a vibrating particle from equilibrium position is represented is called phase angle.

(viii) Epoch (ϕ) = The phase angle at time $t=0$ (Initially) is called epoch.

(ix) phase difference = The difference between the phase angles of two particles moving in simple harmonic motion from equilibrium position is

Called phase difference

⇒ Frequency = $f = \frac{1}{T}$
Unit = Hertz (Hz) or Sec^{-1}

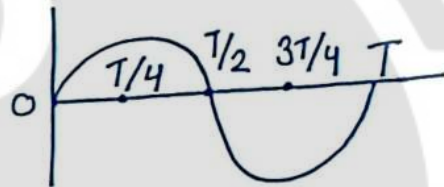
⇒ Angular frequency =

$$\omega = \frac{2\pi}{T}$$

$$\boxed{\omega = 2\pi f}$$

Unit = radian per Sec (rps)

⇒ Graphical representation of S.H.M. = $y = A \sin \omega t$



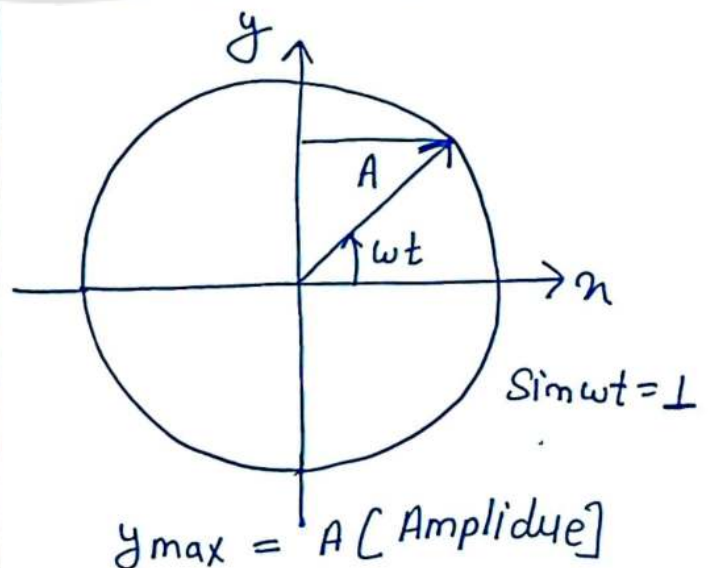
⇒ Simple Harmonic motion

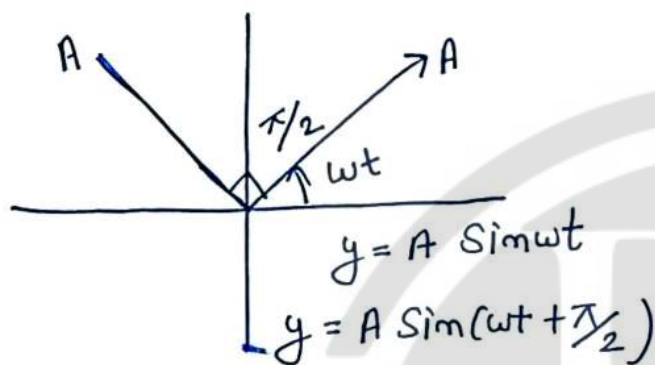
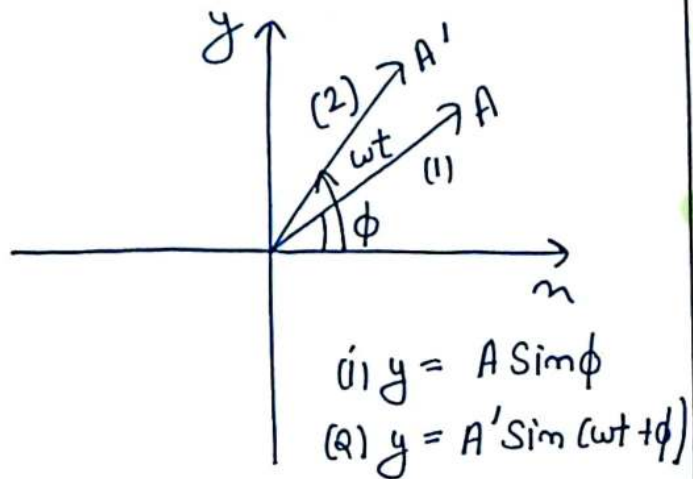
(motion of projection of uniform circular motion)

→ phase diagram

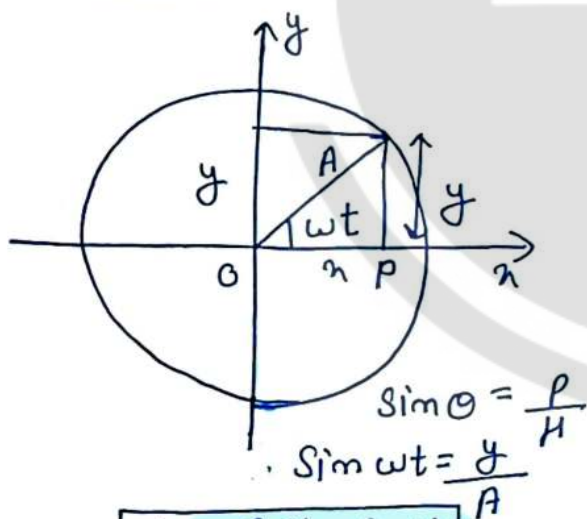
Displacement of particle in S.H.M.

$$y = A \sin(\omega t)$$

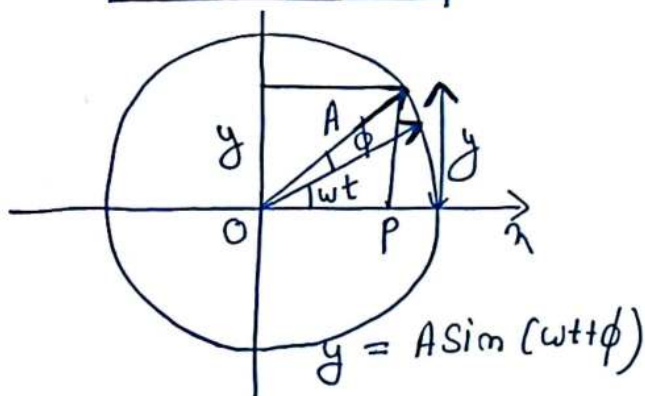




⇒ Equation of displacement by phasor =



$$y = A \sin (\omega t)$$



⇒ Graphical representation of displacement, S.H.M. =

→ displacement in S.H.M. =

Let the displacement of particle in S.H.M. at $t = 0$

$$y = A \sin \omega t$$

Let initially epoch $\phi = 0$ and angular frequency

$$\omega = \frac{2\pi}{T}$$

→ Then displacement

$$y = A \sin \frac{2\pi}{T} t$$

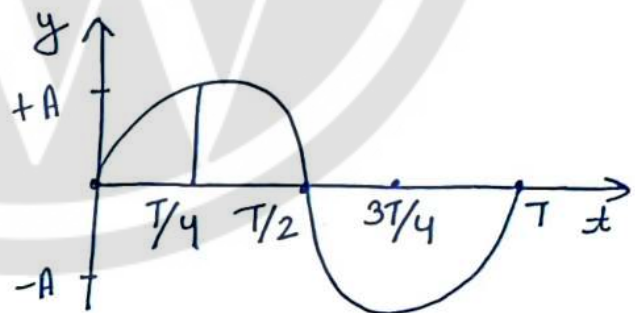
$$y = A \sin \frac{2\pi}{T} t$$

① at $t = 0$

$$y = A \sin \frac{2\pi}{T} \times 0$$

$$y = A \sin 0 \Rightarrow y = 0$$

→ displacement time graph



⇒ Velocity of S.H.M. = Let

The displacement at any instant

$$y = A \sin \omega t$$

$$\text{Velocity} = \frac{dy}{dt}$$

$$v = \frac{dA \sin \omega t}{dt}$$

$$V = A \frac{d}{dt} \sin \omega t$$

$$V = A \omega \cos \omega t$$

$$V = A \omega \cos \omega t$$

$$[\cos \omega t = \sqrt{1 - \sin^2 \omega t}]$$

$$V = A \omega \sqrt{1 - \sin^2 \omega t}$$

$$V = \omega \sqrt{A^2 (1 - \sin^2 \omega t)}$$

$$V = \omega \sqrt{A^2 - A^2 \sin^2 \omega t}$$

$$V = \omega \sqrt{A^2 - y^2} \quad [y = A \sin \omega t]$$

①

$$V = A \omega \cos \omega t$$

$$V = \omega \sqrt{A^2 - y^2}$$

Case ① When $y = 0$
mean position

$$V = \omega \sqrt{A^2 - 0}$$

$$V = \omega A \text{ max}$$

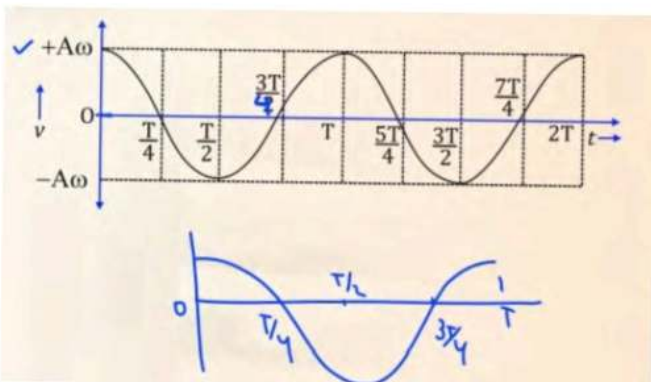
Case ② = at Extreme position

$$y = A$$

$$V = \omega \sqrt{A^2 - A^2}$$

$$V = 0$$

⇒ Graph between velocity and time =



t	$\omega t = \frac{2\pi}{T}t$	$\cos \omega t$	$v = A\omega \cos \omega t$
0	$\frac{2\pi}{T} \cdot 0 = 0$	$\cos 0 = 1$	$+A\omega$
$\frac{T}{4}$	$\frac{2\pi}{T} \cdot \frac{T}{4} = \frac{\pi}{2}$	$\cos \frac{\pi}{2} = 0$	0
$\frac{T}{2}$	$\frac{2\pi}{T} \cdot \frac{T}{2} = \pi$	$\cos \pi = -1$	$-A\omega$
$\frac{3T}{4}$	$\frac{2\pi}{T} \cdot \frac{3T}{4} = \frac{3\pi}{2}$	$\cos \frac{3\pi}{2} = 0$	0
T	$\frac{2\pi}{T} \cdot T = 2\pi$	$\cos 2\pi = 1$	$+A\omega$

⇒ Graphical representation of velocity of S.H.M. =

② Graph B/w velocity and displacement =

⇒ Equation of displacement in S.H.M.

$$y = A \sin \omega t$$

$$\frac{y}{A} = \sin \omega t \text{ --- ①}$$

⇒ Equation of velocity in S.H.M.

$$V = A \omega \cos \omega t$$

$$\frac{V}{A\omega} = \cos \omega t \text{ --- ②}$$

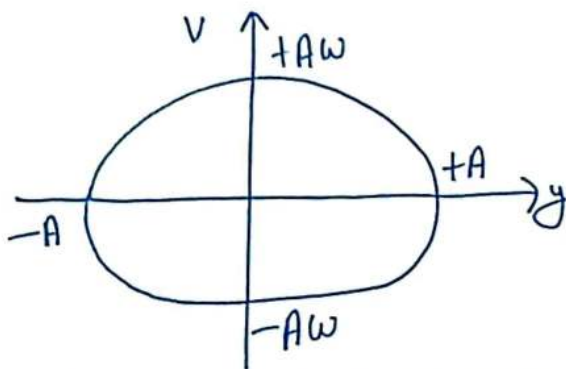
$$\text{①}^2 + \text{②}^2$$

$$\frac{y^2}{A^2} + \frac{V^2}{(A\omega)^2} = \sin^2 \omega t + \cos^2 \omega t$$

$$\frac{y^2}{A^2} + \frac{V^2}{(A\omega)^2} = 1$$

So graph B/w v and displacement will be ellipse

Graph between v and y =



⇒ Acceleration in S.H.M =

Velocity at any instant in S.H.M

$$v = Aw \cos \omega t$$

⇒ acceleration

$$a = \frac{dv}{dt}$$

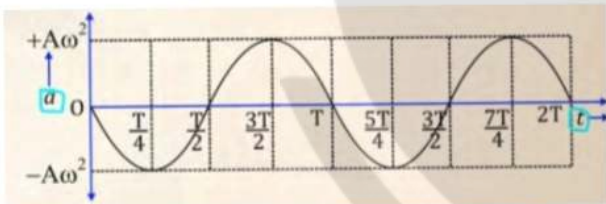
$$a = \frac{d}{dt} Aw \cos \omega t$$

$$a = Aw \frac{d}{dt} \cos \omega t$$

$$a = Aw [-\omega \sin \omega t]$$

$$a = -\omega^2 y$$

$$a \propto -y$$

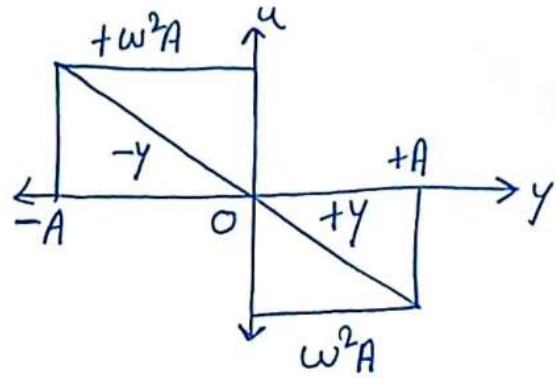


t	$\omega t = \frac{2\pi}{T} t$	$\sin = \omega t$	$a = -A\omega^2 \sin \omega t$
0	$\frac{2\pi}{T} \cdot 0 = 0$	$\sin 0 = 0$	0
$\frac{T}{4}$	$\frac{2\pi}{T} \cdot \frac{T}{4} = \frac{\pi}{2}$	$\sin \frac{\pi}{2} = 1$	$-A\omega^2$
$\frac{T}{2}$	$\frac{2\pi}{T} \cdot \frac{T}{2} = \pi$	$\sin \pi = 0$	0
$\frac{3T}{4}$	$\frac{2\pi}{T} \cdot \frac{3T}{4} = \frac{3\pi}{2}$	$\sin \frac{3\pi}{2} = -1$	$+A\omega^2$
T	$\frac{2\pi}{T} \cdot T = 2\pi$	$\sin 2\pi = 0$	0

⇒ Graph Between acceleration and displacement =

$$a = -\omega^2 y$$

$$a \propto -y$$



Example = On an average a human heart is found to beat 75 times in a minute. Calculate its frequency and period.

$$\text{Ans} = f = 75 \text{ time/minute}$$

$$f = \frac{75}{60} = 1.25 \text{ per sec or Hz}$$

$$T = \frac{1}{f}$$

$$T = \frac{1}{1.25} = 0.8 \text{ Sec}$$

② A body is in S.H.M of 1 m amplitude. its velocity at mean position is 10 m/s. Find its frequency.

$$\text{Ans} = A = 1 \text{ m}$$

$$v_{\max} = 10 \text{ m/s}$$

$$f = ?$$

$$v_{\max} = Aw$$

$$v_{\max} = A(2\pi f)$$

$$f = \frac{v_{\max}}{2\pi A}$$

$$f = \frac{10}{2 \times 3.14 \times 1}$$

$$f = \frac{10}{6.28}$$

$$f = 1.59 \text{ Hz}$$

⇒ Energy of Simple Harmonic

Oscillator = The particle or

Body in Simple harmonic

motion is called Simple

Harmonic oscillator The energy

of Simple Harmonic

Oscillator is of two types =

- (i) Kinetic energy
- (ii) Potential energy
- (iii) Total energy = K.E. + P.E.

⇒ Kinetic energy of a Simple Harmonic oscillator =

Instantaneous Kinetic energy. The Kinetic energy of Simple Harmonic oscillator is present due to the motion of the oscillator

→ Let the velocity of an oscillator at time 't'

$$v = A\omega \cos \omega t \quad \text{--- ①}$$

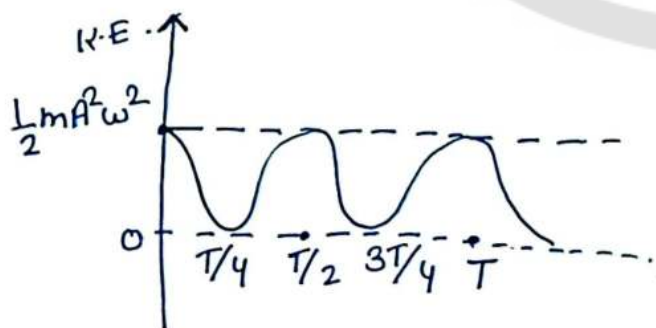
→ A is the Amplitude of oscillator and m is the mass.

The K.E.

$$K.E. = \frac{1}{2}mv^2$$

$$K.E. = \frac{1}{2}mA^2\omega^2 \cos^2 \omega t$$

Graph B/w K.E. and t =



② Velocity in terms of displacement

$$v = \omega \sqrt{A^2 - y^2}$$

Kinetic energy

$$K.E. = \frac{1}{2}mv^2$$

$$K.E. = \frac{1}{2}m\omega^2(A^2 - y^2)$$

① At mean position $y = 0$

$$K.E. = \frac{1}{2}m\omega^2[A^2 - 0]$$

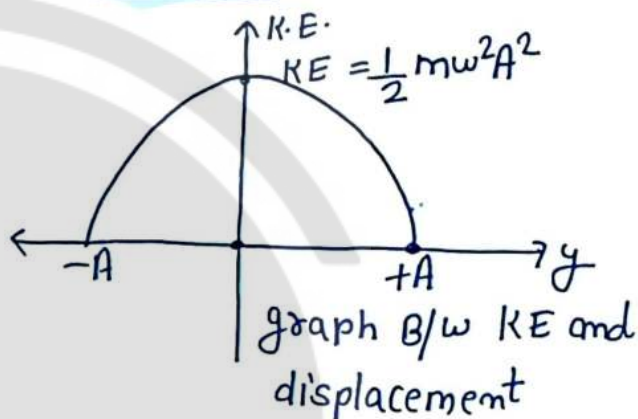
$$K.E. = \frac{1}{2}m\omega^2 A^2$$

maximum value

② At extreme position $y = A$

$$K.E. = \frac{1}{2}m\omega^2[A^2 - A^2]$$

$$K.E. = 0$$



Average Kinetic Energy =

$$\overline{K.E.} = \frac{\int_0^T K.E. dt}{T}$$

$$K.E. = \frac{1}{2}mA^2\omega^2 \cos^2 \omega t$$

$$\overline{K.E.} = \frac{\int_0^T \frac{1}{2}m\omega^2 A^2 \cos^2 \omega t dt}{T}$$

$$\overline{K.E.} = \frac{1}{2} \frac{m\omega^2 A^2}{T} \int_0^T \cos^2 \omega t dt$$

$$\overline{K.E.} = \frac{1}{2} \frac{m\omega^2 A^2}{T} \frac{T}{2} \left| \int_0^T \cos^2 \omega t dt = \frac{T}{2} \right|$$

$$\overline{K.E.} = \frac{1}{4}m\omega^2 A^2$$

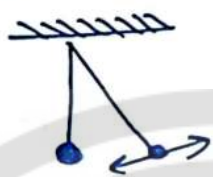
⇒ Potential energy of a Simple Harmonic oscillator =

→ work done to bring an oscillator from mean position will store in the form of potential energy of oscillator in SHM.

→ Instant displacement
 $y = A \sin \omega t$

→ acceleration $a = -\omega^2 y$

→ force $F = ma$
 $F = m(-\omega^2 y)$
 $F = -m\omega^2 y$ — ①



→ work done to displace oscillator at dy distance

$$dw = \vec{F} \cdot d\vec{y}$$

$$dw = F dy \cos 180^\circ$$

$$dw = -F dy$$

put the value of F

$$dw = -(-m\omega^2 y) dy$$

$$dw = +m\omega^2 y dy$$

net work done to displace 'y'

$$w = \int_0^y m\omega^2 y dy$$

$$w = m\omega^2 \int_0^y y dy$$

$$w = m\omega^2 \left[\frac{y^2}{2} \right]_0^y$$

$$w = m\omega^2 \left[\frac{y^2}{2} - \frac{0^2}{2} \right]$$

$$w = \frac{m\omega^2 y^2}{2}$$

→ work done will store in the form of potential energy

$$U = \frac{1}{2} m \omega^2 y^2$$

Graph B/w displacement and potential energy =

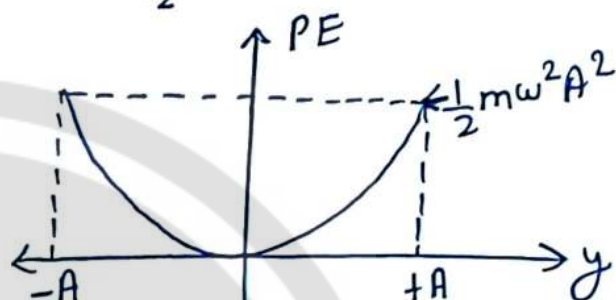
$$U = \frac{1}{2} m \omega^2 y^2$$

① at mean position ($y=0$)

$$U = 0$$

② at extreme position ($y=A$)

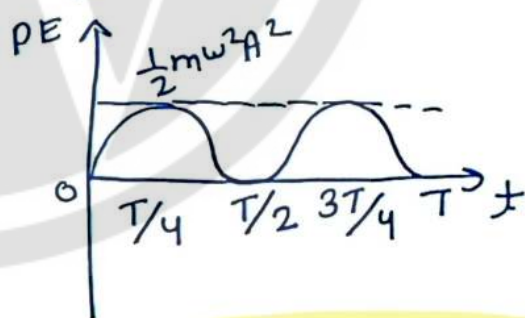
$$U = \frac{1}{2} m \omega^2 A^2 \text{ [max]}$$



→ Potential energy in terms of time =

$$U = \frac{1}{2} m \omega^2 y^2$$

$$U = \frac{1}{2} m \omega^2 A^2 \sin^2 \omega t \quad [y = A \sin \omega t]$$



Average potential Energy =

$$\bar{U} = \frac{1}{T} \int_0^T U dt \text{ — ①}$$

$$U = \frac{1}{2} m \omega^2 A^2 \sin^2 \omega t$$

$$\bar{U} = \frac{1}{T} \int_0^T \frac{1}{2} m \omega^2 A^2 \sin^2 \omega t dt$$

$$\bar{U} = \frac{1}{2} \frac{m \omega^2 A^2}{T} \int_0^T \sin^2 \omega t dt$$

$$\bar{U} = \frac{1}{2} \frac{m\omega^2 A^2}{T} \cdot \frac{T}{2} \int_0^T \sin^2 \omega t dt = \frac{T}{2}$$

$$\bar{U} = \frac{1}{4} m\omega^2 A^2$$

Total Energy of Simple Harmonic Oscillator =

Total Energy is the sum of KE and potential energy

→ Kinetic energy of S.H.M

$$KE = \frac{1}{2} m\omega^2 A^2 \cos^2 \omega t$$

→ potential energy of SHM

$$PE = \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t$$

→ Total energy =

$$E = KE + PE$$

$$E = \frac{1}{2} m\omega^2 A^2 \cos^2 \omega t + \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t$$

$$E = \frac{1}{2} m\omega^2 A^2 [\cos^2 \omega t + \sin^2 \omega t]$$

$$E = \frac{1}{2} m\omega^2 A^2$$

or

$$K.E. = \frac{1}{2} m\omega^2 (A^2 - y^2)$$

$$U = \frac{1}{2} m\omega^2 y^2$$

Total energy

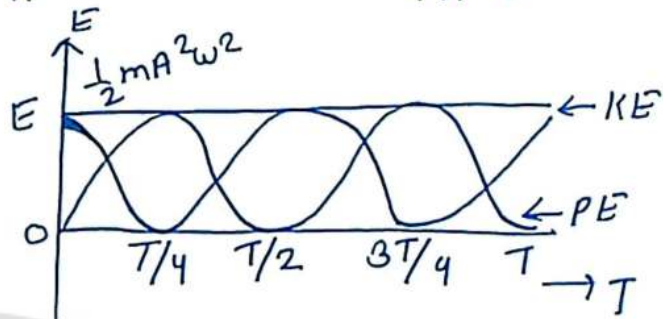
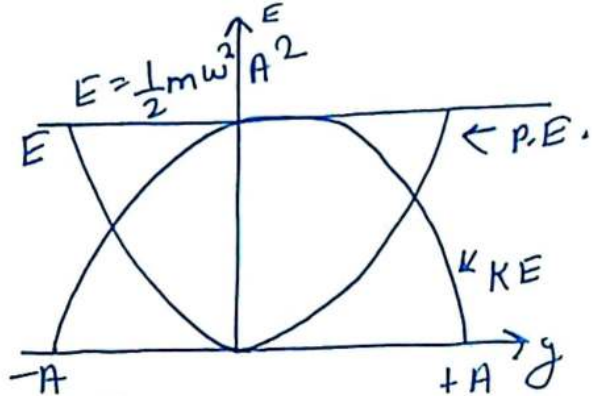
$$E = KE + PE$$

$$E = \frac{1}{2} m\omega^2 (A^2 - y^2) + \frac{1}{2} m\omega^2 y^2$$

$$E = \frac{1}{2} m\omega^2 A^2 - \frac{1}{2} m\omega^2 y^2 + \frac{1}{2} m\omega^2 y^2$$

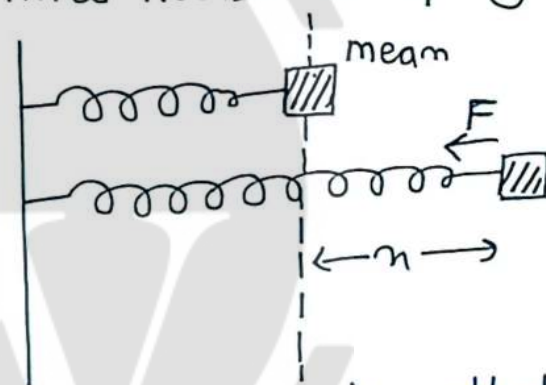
$$E = \frac{1}{2} m\omega^2 A^2$$

Graph of total Energy in S.H.M.



Example of SHM =

① Oscillation of mass attached with a horizontal spring.



→ Let a mass m is attached at one end of spring. The Spring constant is k after displace it by x distance there will be a restoring force.

$$F \propto x$$

$$F = -Kx \text{ --- (1)}$$

→ acceleration

$$a = \frac{F}{m}$$

$$a = \frac{-Kx}{m} \left[\omega^2 = \frac{K}{m} \right]$$

$$a = -\omega^2 x$$

$$a \propto -x$$

○ve sign indicate the direction towards equilibrium position

⇒ Time period of SHM

$$T = \frac{2\pi}{\omega} \left[\omega = \sqrt{\frac{k}{m}} \right]$$

$$T = \frac{2\pi}{\sqrt{k/m}}$$

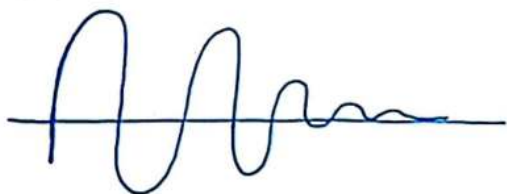
$$T = 2\pi \sqrt{\frac{m}{k}} \quad \left[\begin{array}{l} \text{more mass} \\ T \uparrow \uparrow \end{array} \right]$$

Free oscillations = In the absence of external force the oscillations of the object are called free oscillations.

$$F_{\text{ext}} = 0$$

Damped oscillations = The oscillations of an object in which the amplitude decrease with time and becomes zero are called damped oscillations.

→ air resistance



Example = A 5 Kg collar is attached to a spring of spring constant 500 N m^{-1} . It slides without friction over a horizontal rod. The

Collar is displaced from its equilibrium position by 10.0 cm and released.

- (a) The period of oscillation
(b) The maximum speed and
(c) maximum acceleration of the collar $m = 5 \text{ Kg}$, $k = 500 \text{ N/m}$,
 $A = 10 \text{ cm}$, $A = 10 \times 10^{-2}$, $A = 10^{-1}$
 $A = 0.1 \text{ m}$

→ maximum acceleration

$$a = \omega^2 A \quad \left[\omega^2 = \frac{k}{m} \right]$$

$$a = \frac{k}{m} A$$

$$a = \frac{500 \times 0.1}{5}$$

$$a = 100 \times 0.1$$

$$a = 10 \text{ m/s}^2$$

$$\textcircled{1} T = 2\pi \sqrt{\frac{m}{k}}$$

$$T = 2\pi \sqrt{\frac{5}{500}}$$

$$T = 2\pi \sqrt{\frac{1}{100}}$$

$$T = 2 \times 3.14 \times \frac{1}{10}$$

$$T = \frac{6.28}{10} =$$

$$0.63 \text{ Sec}$$

$$\textcircled{2} V_{\text{max}} = A\omega$$

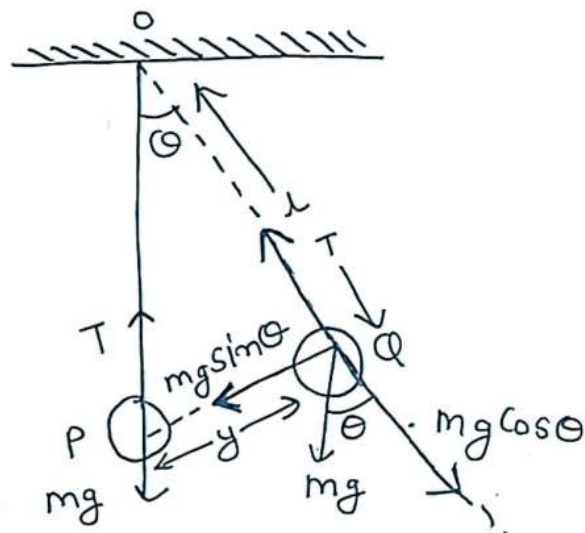
$$V_{\text{max}} = A \sqrt{\frac{k}{m}} \quad \left[\omega = \sqrt{\frac{k}{m}} \right]$$

$$V_{\text{max}} = 0.1 \sqrt{\frac{500}{5}}$$

$$V_{\text{max}} = 0.1 \times 10$$

$$V_{\text{max}} = 1 \text{ m/s}$$

Simple pendulum =



→ A point mass of m is tied with one end of weightless Thread whose length does not change and it is suspended with other end to any rigid surface. This arrangement is called simple pendulum.

→ Now the Bob of simple pendulum is displaced by an angle θ from mean position

→ There are two forces on the Bob.

① Gravitational force

$$F = mg \text{ (downwards)}$$

② Tension force

T [Toward suspension point]

→ Tension is balanced by the component of $mg \cos \theta$ and the Bob will move towards equilibrium position due to $mg \sin \theta$.

⇒ Restoring force

$$F = -mg \sin \theta$$

[θ is very small so $\sin \theta \approx \theta$]

$$\Rightarrow F = -mg\theta \text{ --- ①}$$

If a is acceleration then

$$F = ma \text{ --- ②}$$

$$ma = -mg\theta$$

$$a = -g\theta \text{ --- ③}$$

$$a = -\omega^2 y$$

Let displacement is y

$$a = -\omega^2 y$$

In $\triangle OPQ$, for small angle

$$\theta = \frac{y}{L}$$

put the value of θ and a

in equation ③

$$-\omega^2 y = -g \frac{y}{L} \quad \left[\omega = \frac{2\pi}{T} \right]$$

$$\left(\frac{2\pi}{T} \right)^2 = \frac{g}{L}$$

$$\frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Time period does not depend on mass

→ Time period depends on

① Length [effective length]

$$T \propto \sqrt{L}$$

$$T \propto \frac{1}{\sqrt{g}}$$

⇒ If solid sphere is replaced by hollow sphere then what will be effect of time if mass is not change



Second's pendulum = The pendulum whose time period is 2 sec. is called second pendulum

$$T = 2 \text{ Sec}$$

Example = what is the length of a simple pendulum which ticks after each second.

$$\text{Ans} = T = 2\pi \sqrt{\frac{L}{g}}$$

$$\frac{T}{2\pi} = \sqrt{\frac{L}{g}}$$

$$\frac{T^2}{(2\pi)^2} = \frac{L}{g}$$

$$L = \frac{T^2 g}{4\pi^2}$$

$$L = \frac{2 \times 2 \times 9.8}{4 \times 3.14 \times 3.14}$$

$$L = 0.99$$

$$L \approx 1 \text{ m}$$